# NexusIQ Corporation — Inventory Reorder Standard Operating Procedure

Operations SOP | Version 2.1 | Effective: March 1, 2024

Owner: Supply Chain Operations | Review cycle: Quarterly | Last revised: February 20, 2024 | Classification: Internal

## 1. Purpose and Scope

This SOP defines procedures for inventory monitoring, reorder threshold triggers, purchase order initiation, and emergency procurement across all NexusIQ distribution facilities. The inventory table covers 2,000 records representing 100 stores across five regions — Central, East, West, South, and North — each stocking 20 product SKUs across the Electronics, Home, Sports, Clothing, and Food categories.

As of the latest inventory snapshot, 72 SKUs across the network are below their defined reorder point, representing 3.6% of total inventory positions. This figure is above the target threshold of 2.5% and has triggered a Category 2 supply chain alert per Section 5 of this document. Electronics and Home categories account for the majority of below-reorder positions, driven by Q4 holiday demand exceeding forecast by 14% in Electronics and 9% in Home.

This document should be read alongside the Seasonal Demand Incident Report (November 2024) and the Inventory Shortage Root Cause Analysis, which provide background on the Q4 demand surge that produced current low-stock conditions.

## 2. Reorder Threshold Definitions

Reorder points are calculated using a standard formula: Reorder Point = (Average Daily Sales x Lead Time Days) + Safety Stock. Safety stock is set at 1.5 standard deviations of daily demand variability to provide approximately 93% service level protection. Lead time assumptions are reviewed quarterly with each vendor and updated in the procurement system. Electronics carries the longest lead time due to international component sourcing; Food carries the shortest due to regional supplier networks.

| Category | Avg Daily Units Sold | Lead Time (days) | Safety Stock (units) | Reorder Point | Current Low-Stock SKU |
|---|---|---|---|---|---|
| Electronics | 42 units | 21 days | 88 units | 970 units | 28 SKUs |
| Home | 38 units | 18 days | 72 units | 756 units | 19 SKUs |
| Sports | 31 units | 14 days | 58 units | 492 units | 12 SKUs |
| Clothing | 29 units | 16 days | 55 units | 519 units | 8 SKUs |
| Food | 24 units | 7 days | 45 units | 213 units | 5 SKUs |
| All Categories | 164 units avg | 15 days avg | 64 units avg | 590 units avg | 72 SKUs total |

*Source: inventory table, procurement system. As of latest snapshot. Lead times per vendor agreements.*

## 3. Reorder Initiation Procedures

When a SKU falls below its reorder point, the inventory management system generates an automated purchase order draft within four hours. The draft is reviewed by the category buyer before submission to avoid duplicate orders or orders that conflict with pending inbound shipments. For standard reorders the buyer review window is 24 hours; for high-velocity SKUs — defined as those with daily sales exceeding twice the category average — the window is reduced to four hours.

Purchase orders are submitted electronically to primary vendors via the supplier portal. Vendors are required to acknowledge receipt within 8 business hours and provide a confirmed ship date within 24 hours. Orders not acknowledged within the window are escalated to the supply chain manager and the backup vendor is contacted simultaneously.

Quantities ordered follow an Economic Order Quantity calculation to balance ordering costs against carrying costs. In practice, EOQ quantities are rounded up to the nearest pallet quantity to optimize freight costs. During Q4 seasonal periods (October through December), order quantities are multiplied by a 1.3 seasonal factor for Electronics and Home categories based on historical demand patterns.

## 4. Current Low-Stock Alert Summary

The following stores have the highest concentration of below-reorder-point SKUs as of the latest inventory snapshot. Store identifiers follow the format [Region Initial][Store Number]. Stores with five or more low-stock positions are classified as high-priority and receive daily replenishment review until stock levels are restored. The current situation is concentrated in West and Central region stores, consistent with their higher Electronics transaction mix as documented in the Regional Performance Analysis.

| Store ID | Region | Low-Stock SKUs | Most Affected Category | Expected Restock Date | Priority |
|---|---|---|---|---|---|
| W014 | West | 8 SKUs | Electronics | Dec 18, 2024 | HIGH |
| W007 | West | 7 SKUs | Electronics, Home | Dec 19, 2024 | HIGH |
| C003 | Central | 6 SKUs | Electronics | Dec 20, 2024 | HIGH |
| W019 | West | 5 SKUs | Home | Dec 21, 2024 | HIGH |
| C011 | Central | 5 SKUs | Electronics | Dec 21, 2024 | HIGH |
| E008 | East | 4 SKUs | Home, Sports | Dec 22, 2024 | MEDIUM |
| S002 | South | 4 SKUs | Clothing | Dec 22, 2024 | MEDIUM |
| N005 | North | 3 SKUs | Sports | Dec 24, 2024 | NORMAL |
| E015 | East | 3 SKUs | Electronics | Dec 24, 2024 | NORMAL |
| All others | Various | 27 SKUs (2 each) | Mixed | Dec 22–28, 2024 | NORMAL |

*72 total SKUs below reorder point as of snapshot. 26 HIGH priority stores require daily review.*

## 5. Emergency Reorder Procedures

When a store reaches zero inventory on a high-velocity SKU before a scheduled reorder arrives — a stockout event — the supply chain team initiates an emergency transfer from the nearest regional distribution center with available stock. Emergency transfers carry a 40% premium over standard replenishment costs due to expedited freight. In FY 2024, 23 emergency transfers were executed, predominantly in Q4; total emergency freight premium cost was approximately $87,400.

If regional distribution center stock is also depleted, the category buyer contacts the top three vendors for emergency spot orders. Spot order pricing is typically 8-15% above contracted pricing. Any spot order exceeding $25,000 requires Supply Chain Manager approval. The decision to accept spot order pricing versus accepting a stockout period is made on a SKU-by-SKU basis considering the product margin, expected stockout duration, and availability of substitutes.

Following any emergency reorder event, a root cause analysis is completed within five business days. The analysis examines whether the event resulted from forecast error, vendor delivery failure, or demand anomaly. Results feed into the quarterly reorder parameter review. For the most recent emergency reorder analysis, see the Inventory Shortage Root Cause Analysis document.

## 6. Vendor Performance Monitoring

Vendors are scored quarterly on three dimensions: on-time delivery rate (target >95%), fill rate on ordered quantities (target >98%), and product quality acceptance rate (target >99.5%). Vendors falling below target on any dimension receive a formal performance notice and must submit a corrective action plan within 14 days. Vendors with two consecutive quarters below target on delivery or fill rate are placed on probation and the backup vendor share is increased from 15% to 40% of category orders.

| Category | Primary Vendor | On-Time Delivery | Fill Rate | Quality Rate | Status |
|---|---|---|---|---|---|
| Electronics | TechSource Global | 94.2% | 97.8% | 99.6% | Performance Notice — Delivery |
| Home | HomeSupply Co | 96.1% | 98.4% | 99.7% | Good Standing |
| Sports | SportPro Dist. | 95.8% | 98.1% | 99.8% | Good Standing |
| Clothing | FashionLink | 93.7% | 97.2% | 99.4% | Performance Notice — Fill Rate |
| Food | FreshChain Ltd | 97.3% | 99.1% | 99.9% | Good Standing |

*Q4 2024 vendor scorecard. TechSource and FashionLink on performance notice.*